

Assessment Task 2: Design Development and Application (40%).

This assignment was locked Sep 15 at 23:59.

Course code and name: COSC2753 Machine Learning

Length: + Report of no more than 5 pages plus up to 2 pages for appendices (excluding Cover Page, References List), written in font size 12;

+ And source code (including models)

Type: Group of 04-05 students

Feedback mode: Written feedback

Late work: + For assignments 01 to 05 days late, a penalty of 20% (of the marks awarded) per day will apply.

+ For assignments more than 05 days late, a penalty of 100% will apply.

+ Weekend days (Saturday and Sunday) are counted when counting total late days.

Learning Objectives Assessed

- **CLO1:** Understand the fundamental concepts and algorithms of machine learning and applications
- **CLO2:** Understand a range of machine learning methods and the kinds of problem to which they are suited
- **CLO3 :**Set up a machine learning configuration, including processing data and performing feature engineering, for a range of applications
- **CLO4:** Apply machine learning software and toolkits for diverse applications

Ready for Life and Work

Completing this assessment would help you achieve the following employability skills:

- Analyze and model requirements and constraints for the purpose of designing and implementing solutions to a learning challenge
- Evaluate and compare approaches and algorithms on the basis of the nature of the problem/task being addressed
- Interpret abstract theoretical propositions, choose methodologies, justify conclusions and defend professional decisions to both IT and non-IT personnel via technical reports of a professional standard and technical presentations

Assessment Details

In this assignment you will design and create an end-to-end machine learning system for a real-world problem. This assignment is designed for you to apply and practice skills of critical analysis and evaluation to circumstances similar to those found in real-world problems. This is a group project (groups of 4 or 5). In this assignment you will:

- Design and Create an end-to-end machine learning system
- Apply multiple algorithms to a real-world machine learning problem
- Analyze and Evaluate the output of the algorithms
- Research into extending techniques that are taught in class
- Provide an ultimate judgement of the final trained model(s) that you would use in a real-world setting

More details of the Assignment 2 can be downloaded here [COSC2753 2026B Assignment 2.pdf](#)

Dataset

The dataset for Assignment 2 is here [A2_FashionDataset.zip](#) ↓

Negotiated Project template

Template for negotiated project is [here](#).

Support Resources

This assessment requires that you meet RMIT's expectations for academic integrity. More information and advice on how to avoid plagiarism are available in the Getting Started module.

Open [the academic integrity page](#).

Additional library and learning resources are available to help with the assessment in this course

Link to [Assignment Support](#).

Submission Instruction

Only one member of your group should submit the necessary files. Please do not spread your submission across both members of your group. You will need to submit the following deliverables:

- The acceptable formats for the report are PDF, DOC, DOCX. Please name the report by following this convention: *COSC2753_A2_<group id>*, where *<group_id>* is your group id as in the Canvas, such as *S1_G3*, or *H1_G2* or *S3_G4*
 - **Please submit YOUR REPORT in this Canvas page**

- Your Python scripts or Jupyter notebooks used to perform your analysis.
 - Putting three (or more) models and your code (including Jupyter notebook file and the scripts for running your models for predicting) in one folder.
 - Zip your folder before submitting it. Name the zip file by following the same convention above
 - Please make sure that the notebook must be readily opened and executed on any standard machine/setting. The best practice is to include a README file to instruct the user on how to set up the environment to run your file (e.g., put data files in which folder, install which libraries/packages etc.)
 - Please note that your code will be checked for plagiarism by both our specialized software and by Turnitin
 - Please submit your code zip file in the separate Canvas page - CODE SUBMISSION
- Your Prediction file
 - Must use the provided structure file
 - DO NOT CHANGE the format of the provided file
 - Please name the prediction submission by following this convention: **COSC2753_A2_<group_id>.csv**
 - where <group_id> is your group id as in the Canvas, such as S1_G3, or H1_G2 or S3_G4, ...
 - Please submit your code zip file in the separate Canvas page - PREDICTION SUBMISSION

Rubric

You can view the rubric here on this Canvas page

Criteria	Ratings					Pts
Approach	50 to >39.0 pts HD Outstanding across the course. The approach is an excellent and extremely thorough investigation of the chosen ML problem. It explored multiple algorithms and techniques for solving the problem. It goes beyond using the tools provided in class. There are no gaps in what the investigation has considered. The approach makes careful consideration of the unique aspects of the chosen ML problem----- The choices in the investigation are highly justified and have an excellent explanation. The reader is left with no	39 to >30.0 pts DI The approach is a good and reasonably thorough investigation of the chosen ML problem. It is mostly limited to the techniques and algorithms discussed in class. There are small gaps between in the investigation in what could have been explored. The approach considers, but not fully, the unique aspects of the chosen ML problem----- -The choices in the investigation are justified and have a good explanation. The reader is left with minor questions----- The design of the approach includes all necessary steps including the choice of training data	30 to >25.0 pts CR The approach is sufficient, but not a thorough investigation of the chosen ML problem. It is limited to the techniques and algorithms discussed in class, such as a pure reproduction of lab code. There are gaps in the investigation and alternative algorithms or techniques are better than the ones in the approach. The approach has a limited consideration of the unique aspects of the chosen ML problem----- The choices in the investigation have sufficient justification, however, there are unexplained choices and the reader is left with important open	25 to >17.0 pts PA The approach is a minimally sufficient investigation of the chosen ML problem. It only examines the bare minimum requirements of suitable techniques and algorithms. There are many gaps in the investigation and there are algorithms or techniques are clearly more suited to the chosen ML problem. The approach has limited or minimal consideration of the unique aspects of the chosen ML problem----- The choices in the investigation have the bare minimal justification. The reader is left with important unanswered questions that	17 to >0 pts NN Poor, superficial, or incomplete approach that does not meet the minimum requirements for PA----- Does not justify the approach. Does not explain the approach----- -The design of the approach is missing significant steps such as the choice of training data selection, pre-processing, model training, parameter tuning, and evaluation.	50 pts

Criteria	Ratings					Pts
	questions.----- The design of the approach includes excellent choice of training data selection, preprocessing, model training, parameter tuning, and evaluation. Be able to showcase your understanding not only of building the ML model but also of how to integrate ML model into a real-world application.	selection, pre-processing, model training, parameter tuning, and evaluation. Be able to showcase your understanding not only of building the ML model but also of how to integrate ML model into a real-world application.	questions----- The design of the approach includes most of the necessary steps such as the choice of training data selection, preprocessing, model training, parameter tuning, and evaluation.	are not considered----- The design of the approach includes most of the necessary steps such as the choice of training data selection, preprocessing, model training, parameter tuning, and evaluation.		
Ultimate Judgement and Evaluation	30 to >24.0 pts HD Outstanding across the course. Ultimate Judgement is established and exceptionally justified----- Evaluation of the Ultimate Judgement is exceptional and clearly demonstrated (or proves) the viability of the trained model	24 to >20.0 pts DI Ultimate Judgement is established and suitably justified--- Evaluation of the Ultimate Judgement is sound and suitably explained, however, the reader may not be fully convinced and have minor questions---	20 to >15.0 pts CR Ultimate Judgement is established, but there are unexplained choices, or the justification is hard to follow-- ----An sufficient attempt at evaluating the Ultimate Judgement is made-----The evaluation is independent.	15 to >10.0 pts PA An Ultimate Judgement is made by not justified----- The Ultimate Judgement is not evaluated or the evaluation is incomplete and insufficient----- The evaluation is not independent.	10 to >0 pts NN Not completed. An Ultimate Judgement is not made.	30 pts

Criteria	Ratings					Pts
	in real-world practice----- The evaluation is independent.	The evaluation is independent.				
Report Presentation	20 to >17.0 pts HD Outstanding across the course. Report is easy to read and flows well. It is structured well, leading the reader to fully understand the rationale for the final approach taken----- Approaches are excellently described. Tables, figures and other visualisation are tailored to the descriptions and justifications made in the report's text.	17 to >12.0 pts DI Report is reasonably easy to read and flows relatively well. It is structured reasonably well, leading the reader to reasonably understand the rationale for the final approach taken----- Approaches are described well, Tables, figures and other visualisation are easy to read and to interpret, but may not be highly relevant or appropriately tailored to the descriptions and justifications in the report's text.	12 to >9.0 pts CR Report can be followed but does not flows well in places. It is adequately structured, but reader may find it difficult to understand the rationale of selected approach----- Approaches are described in sufficient detail, but aspects may be incomplete. Tables, figures and other visualisation are sufficient but are difficult to interpret.	9 to >6.0 pts PA Report is difficult to follow and doesn't flows well. Readers find it difficult to understand the rationale of the selected approach----- Approaches are described to a minimal standard. Tables, figures and other visualisation are provided to a care minimum amount.	6 to >0 pts NN Not completed. Incomplete or error ridden report.	20 pts
Total Points: 100						